

Enhancing water level prediction of the Yangtze River based on RGL-stacking model

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ABSTRACT

Accurate water level prediction along the Yangtze River is essential for navigation safety, flood prevention, and water resource management. However, the basin's complex hydrological dynamics—driven by seasonal monsoon precipitation, Three Gorges Dam regulation, tributary inflows, and tidal influences—pose significant forecasting challenges. This study proposes the RGL-Stacking model, a heterogeneous ensemble deep learning framework that integrates three structurally distinct recurrent base learners—Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM)—to capture complementary temporal patterns. Each base model independently generates predictions from historical water level sequences, and these predictions are combined as meta-features and processed by a Meta-Multilayer Perceptron (Meta-MLP) to produce the final forecast. To prevent information leakage between the base model training and meta-learner training stages, a time-series-aware cross-validation procedure is employed for out-of-fold meta-feature generation. Using daily water level records (November 2003 to December 2010; 2610 days) from four hydrological stations in the middle and lower reaches of the Yangtze River—Yichang, Shashi, Hankou, and Nanjing—we evaluate the model for one-day-ahead prediction. The RGL-Stacking model achieves Mean Absolute Error (MAE) values of 0.0866, 0.0683, 0.0595, and 0.0563 m, and Nash–Sutcliffe Efficiency (NSE) values of 0.998, 0.999, 0.999, and 0.998 at the four stations, consistently outperforming individual RNN, GRU, LSTM, Support Vector Machine (SVM), and Transformer models. Ablation experiments confirm the positive contribution of each base learner, and residual correlation analysis verifies their complementarity. Separate evaluations during the dry season and flood season demonstrate that the RGL-Stacking model maintains superior accuracy under both low-flow and high-flow conditions. Compared with existing hybrid deep learning approaches, RGL-Stacking differs fundamentally in its parallel heterogeneous architecture, fully independent base model training, nonlinear meta-learning combination, and time-series-aware cross-validation for information leakage prevention—addressing key limitations of serial hybrid models (e.g., CNN-LSTM) and homogeneous ensembles. The model achieves these improvements with only modest computational overhead (approximately 2.1 times the parallel training time of a single LSTM, 0.05 s inference per station), demonstrating practical feasibility for real-time deployment in both the Yangtze River navigation channel management system and operational flood early warning platforms.

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1. Introduction

1.1. Background and motivation

Global climate change has intensified the frequency and severity of extreme hydrological events, making accurate water resource management and timely flood early warning increasingly critical challenges [1]. The Yangtze River—one of Asia's largest rivers, spanning approximately 6300 km in length and draining an area of $1.8 \times 10^6 \text{ km}^2$ —plays a central role in the socio-economic development of China. In the context of ongoing digital waterway construction, accurate water level prediction along the Yangtze River is a fundamental requirement for water resource management, navigation safety, and flood control [2].

However, accurate water level forecasting at stations along the middle and lower reaches of the Yangtze River is inherently challenging. This difficulty arises from the complex interplay of multiple drivers: seasonal monsoon variability, spatially distributed rainfall patterns, regulated discharge from the Three Gorges Dam, diverse channel topography, tributary confluence dynamics, and tidal backwater effects in the lower reaches [3]. These factors collectively produce highly nonlinear and non-stationary water level dynamics that traditional statistical approaches struggle to represent. For example, autoregressive integrated moving average (ARIMA) models [4] and their seasonal variants assume linear dependence structures and stationarity, assumptions that are routinely violated in Yangtze River hydrology [5].

1.2. Related work

Machine learning and deep learning methods have substantially advanced the state of water level forecasting over the past decade [6–9]. Among machine learning approaches, Nguyen et al. [10] demonstrated that support vector machines (SVM) can achieve reliable performance for river water level prediction. Khalaf et al. [11] combined Internet of Things (IoT) sensor networks with machine learning classifiers to enhance real-time flood severity assessment. Seo et al. [12] integrated wavelet decomposition with neural networks and fuzzy logic, showing improved forecasting accuracy on daily water level data from South Korean stations.

Recurrent neural networks, particularly long short-term memory (LSTM) networks, have become a dominant approach for hydrological time series forecasting due to their ability to capture temporal dependencies across multiple scales. Widiyari et al. [13] applied LSTM to river water level prediction using upstream and downstream rainfall and water level inputs. Liang et al. [14] demonstrated that LSTM outperforms SVM for daily water level prediction at Dongting Lake. Ruma et al. [15] used Particle Swarm Optimization (PSO) to tune LSTM hyperparameters for water level forecasting across Bangladesh's river network, evaluating performance using NSE, RMSE, and MAE. Granata and Di Nunno [16] applied Bayesian optimization to systematically compare MLP, RBF-NN, LSTM, and Bi-LSTM for short- and medium-term streamflow forecasting. Di Nunno et al. [17] further showed that an ensemble combining deep learning models with MLP and Random Forest can improve short-term streamflow accuracy beyond any single model.

More recently, attention-based architectures have expanded the toolkit for hydrological forecasting. The Temporal Fusion Transformer (TFT) [18] uses variable selection networks and multi-head self-attention to handle multi-horizon prediction with interpretable feature importance. The Informer [19] addresses the quadratic complexity of standard Transformers through sparse attention, enabling efficient long-sequence forecasting. Hybrid architectures that combine convolutional neural networks (CNNs) with recurrent networks (e.g., CNN-LSTM) have also been widely adopted, as CNNs can extract local temporal patterns while recurrent layers model longer-range dependencies [20,21]. Wang et al. [22] incorporated a spatio-temporal attention mechanism into LSTM to improve medium-to long-term

water level prediction along river networks.

1.3. Ensemble methods in hydrological prediction

Ensemble learning improves prediction robustness by combining multiple models, thereby reducing the bias and variance associated with any single model [23]. In hydrological forecasting, ensemble approaches have been developed along three main paradigms.

Homogeneous ensembles aggregate multiple instances of the same model architecture through bagging or boosting [24]. While effective at reducing variance, these methods provide limited diversity because all base learners share the same architectural form and therefore tend to make correlated errors.

Serial hybrid models connect structurally different modules in a fixed sequential pipeline. A common example is wavelet decomposition followed by a neural network predictor [25], where the decomposition step extracts multi-scale features that are then passed to the predictor. However, this design couples the two stages tightly: errors introduced by the feature extraction stage propagate directly to the prediction stage and cannot be corrected downstream.

Stacking ensembles, originally proposed by Wolpert [26] and formalized by Breiman [27], offer a more flexible alternative. Rather than fixing the combination rule, stacking trains a meta-learner to discover the optimal way to combine predictions from multiple diverse base models. Granata et al. [28] demonstrated the potential of this approach by stacking traditional machine learning algorithms with bidirectional LSTM for multi-step streamflow forecasting.

Despite these advances, the use of stacking ensembles that systematically combine structurally diverse recurrent neural network architectures—exploiting their complementary temporal modeling capabilities—remains largely unexplored for water level forecasting along the Yangtze River.

Critically, existing hybrid deep learning approaches for hydrological forecasting can be broadly categorized into three paradigms, each with inherent limitations (1) homogeneous ensembles (e.g., bagging multiple LSTM instances): improve stability but cannot exploit cross-architecture diversity since all members share identical inductive biases (2) serial hybrid models (e.g., CNN-LSTM, wavelet-LSTM); chain heterogeneous modules in a fixed sequential pipeline, thereby creating tight coupling between stages—errors in upstream feature extraction propagate irreversibly to downstream prediction, and the entire architecture must be trained jointly, which constrains individual module specialization; and (3) heterogeneous ensembles with simple aggregation (e.g., averaging or linear weighting of RNN, LSTM, GRU outputs) combine diverse architectures but rely on fixed or manually tuned combination weights that cannot adapt to the complementary error structures of base learners across different hydrological conditions. The proposed RGL-Stacking framework is specifically designed to address all three limitations simultaneously.

1.4. Research gaps and contributions

The preceding review reveals four important gaps in the existing literature: (1) Reliance on single-model architectures. Most deep learning studies for Yangtze River water level prediction use a single recurrent model (e.g., LSTM or GRU alone), without systematically leveraging the complementary strengths of different recurrent architectures. No study has evaluated whether combining RNN, GRU, and LSTM through stacking can produce consistent improvements across stations with distinct hydrological characteristics. (2) Predominance of homogeneous ensembles. Ensemble studies in hydrology most commonly use homogeneous methods (bagging or boosting of the same architecture). The potential of heterogeneous stacking—combining architecturally distinct recurrent networks through a learned meta-learner—remains largely unexplored for water level prediction. (3) Incomplete evaluation frameworks. Many existing studies focus on a

single evaluation metric (e.g., RMSE) and do not separately assess model performance under flood and dry season conditions. The Nash–Sutcliffe Efficiency (NSE), which is the standard benchmark metric in hydrology, is rarely combined with season-specific analysis in a single study. (4) Insufficient discussion of computational feasibility. Computational cost analysis is seldom reported alongside prediction accuracy, despite being a critical factor for operational deployment in real-time monitoring systems such as the Yangtze River navigation channel management network.

- To address these gaps, this study makes the following contributions.
- a) We propose RGL-Stacking, a heterogeneous stacking ensemble that combines RNN, GRU, and LSTM as base learners with an MLP meta-learner, specifically designed for water level prediction at multiple stations with diverse hydrological characteristics.
 - b) We establish a rigorous experimental framework with chronological data splitting, out-of-fold meta-feature generation to prevent information leakage, and comprehensive evaluation using NSE, R^2 , MAE, MSE, and RMSE.
 - c) We provide systematic analysis including base model complementarity verification through residual correlation analysis, ablation experiments, and separate evaluation during dry and flood seasons.
 - d) We discuss computational efficiency and practical implications for operational deployment in the Yangtze River navigation channel management system.

The remainder of this paper is organized as follows: Section 2 describes the study area and station selection. Section 3 details the data processing and model architecture. Section 4 presents the experimental results and discussion. Section 5 provides further discussion on complementarity, computational efficiency, practical implications, and limitations. Section 6 concludes the paper.

2. Study area and station selection

This study examines water level data from four pivotal hydrological stations within the middle and lower reaches of the Yangtze River basin: Yichang, Shashi, Hankou, and Nanjing.

Table 1 summarizes the characteristics of the four stations. These stations were selected to capture the diverse hydrological conditions along the Yangtze River's middle and lower reaches.

- a) Yichang Station, located immediately downstream of the Three Gorges Dam, experiences notable water level fluctuations that are significantly influenced by seasonal variations in upstream inflow and the operational management of the dam. These factors contribute to marked cyclic variations in water levels.
- b) Shashi Station, positioned downstream of Yichang, sees its water levels affected by upstream fluctuations as well as local rainfall patterns. The station's water level characteristics are also shaped by the surrounding topography and landforms, highlighting the complex interplay of geographical and hydrological factors.
- c) Hankou Station, located in Wuhan—one of the Yangtze River basin's largest cities—experiences water level fluctuations intricately connected to the city's hydrological infrastructure, including river navigation, urban drainage systems, and rainfall runoff management.
- d) Nanjing Station, situated in Jiangsu Province at the downstream end of the basin, experiences water level fluctuations influenced by both basin runoff and tidal effects, showcasing distinct tidal patterns. Local rainfall in Jiangsu Province and runoff from tributary rivers further affect the water levels at this station.

The water level fluctuations in the middle and lower reaches of the Yangtze River are influenced by seasonal variations, upstream reservoir management, urban drainage systems, and tidal influences. Among these, discharge alterations from the Three Gorges Reservoir play a

Table 1
Characteristics of the four hydrological stations in the middle and lower Yangtze River.

Station	River Reach	Approximate Longitude (°E)	Approximate Latitude (°N)	Key Hydrological Characteristics
Yichang	Upper–Middle	111.28	30.70	Immediately downstream of Three Gorges Dam; water level strongly regulated by dam operations; marked seasonal cycles
Shashi	Middle	112.24	30.31	Downstream of Yichang; influenced by upstream dam releases and local rainfall; topography shapes water level characteristics
Hankou	Middle	114.31	30.58	Confluence with Han River; major national hydrological station; urban hydrological infrastructure influences drainage and runoff
Nanjing	Lower	118.78	32.07	Tidal section of lower Yangtze; water level influenced by basin runoff, local rainfall, and downstream tidal effects

pivotal role in shaping downstream hydrological conditions. In our model, we account for the time delay between discharge changes at the reservoir and their effects at downstream stations, which allows for more precise prediction and a deeper understanding of the water level response to upstream interventions.

3. Data processing and modeling

3.1. Data description and quality control

This study uses daily average water level records from four hydrological stations along the Yangtze River: Yichang, Shashi, Hankou, and Nanjing. The records span from November 1, 2003, to December 31, 2010, comprising a total of 2610 daily observations per station. Data were retrieved through a Python-based acquisition system connected to the Yangtze River Hydrological Network (<http://www.cjh.com.cn>), which provides access to both real-time and historical hydrological data.

Before model training, three quality control steps were applied to the raw data.

- a) Outlier detection and removal: Water level values that deviated from each station's historical range by more than 20% were flagged as potential sensor errors or recording anomalies. Each flagged value was manually cross-checked against records from neighboring upstream and downstream stations; values confirmed as erroneous were removed.

- b) Gap filling: Missing values arising from sensor downtime or irregular observation intervals were filled by interpolation. Linear interpolation was applied for gaps of up to three consecutive days; cubic spline interpolation was used for longer gaps of four to seven days. Gaps exceeding seven days were not present in the dataset.
- c) Noise smoothing: High-frequency fluctuations unrelated to genuine hydrological dynamics can degrade model training stability. A moving average filter was applied to attenuate such noise while preserving the dominant seasonal and inter-annual trends in the water level series. The window length of the moving average was selected to minimize distortion of flood peak timing (see Fig. 1).

Fig. 2 illustrates the historical water level data curves for the four stations, together with the curves after the smoothing process. The comparison reveals that water level variations at all stations demonstrate distinct periodic trends, underscoring the cyclic nature of the water levels in these sections of the Yangtze River.

3.2. Autocorrelation analysis

To uncover the underlying temporal patterns in the water level data, we performed an autocorrelation analysis to examine the dataset's periodicity and correlation attributes. Autocorrelation quantifies the linear dependence of a time series on its own lagged values and is a standard tool for identifying dominant periodicities in hydrological time series [29].

$$r_k = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^N (x_t - \bar{x})^2}, k = 0, 1, 2, \dots \quad (1)$$

where $r_k \in [-1, 1]$ is the autocorrelation coefficient at lag k , x_t is the observed water level at time step t , x_{t+k} is the observed water level at time step $t + k$, and $\bar{x}$ is the sample mean computed over the entire series.

A value of r_k close to +1 indicates strong positive autocorrelation at lag k , meaning that the water level at time t tends to be numerically similar to the level at time $t - k$ (i.e., the two values rise and fall together). A value close to -1 indicates strong negative autocorrelation, meaning that a high water level at time t corresponds to a low level at time $t - k$, and vice versa. Values near zero indicate no significant linear dependence at that lag. Under the null hypothesis that the series is white noise, the approximate 95% confidence bounds for r_k are $\pm 1.96/\sqrt{N}$ [30].

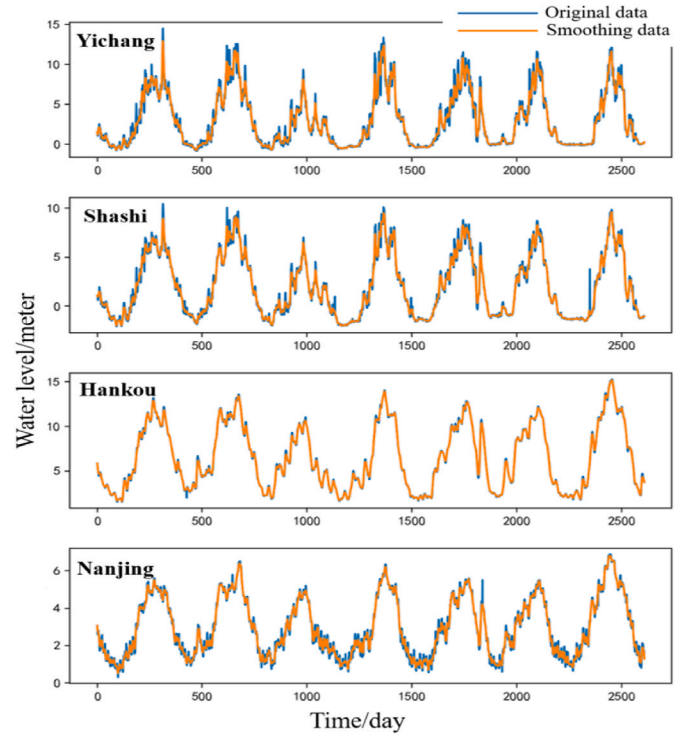


Fig. 2. Original daily water level (blue) and smoothed water level (orange) at four hydrological stations. The x-axis represents time (day), and the y-axis represents normalized water level (m). Obvious annual periodicity can be observed at all stations.

Fig. 3 displays both the water level data and the corresponding autocorrelation coefficient curves for the four stations, where the horizontal axis indicates time lags (in days) and the vertical axis shows the autocorrelation values. Fig. 4 presents bar charts of the autocorrelation coefficients, providing a clearer visualization of the positive and negative correlation structure.

Upon analyzing Figs. 3 and 4, the following patterns are evident across all stations.

- a) Short-lag positive autocorrelation ($k < 60$ days): All stations exhibit high positive autocorrelation, indicating strong day-to-day persistence in water levels driven by sustained hydrological processes.



Fig. 1. Distribution of hydrological stations in the middle and lower reaches of the Yangtze River. The four stations span approximately 1200 km from the Three Gorges Dam exit to the tidal reach.

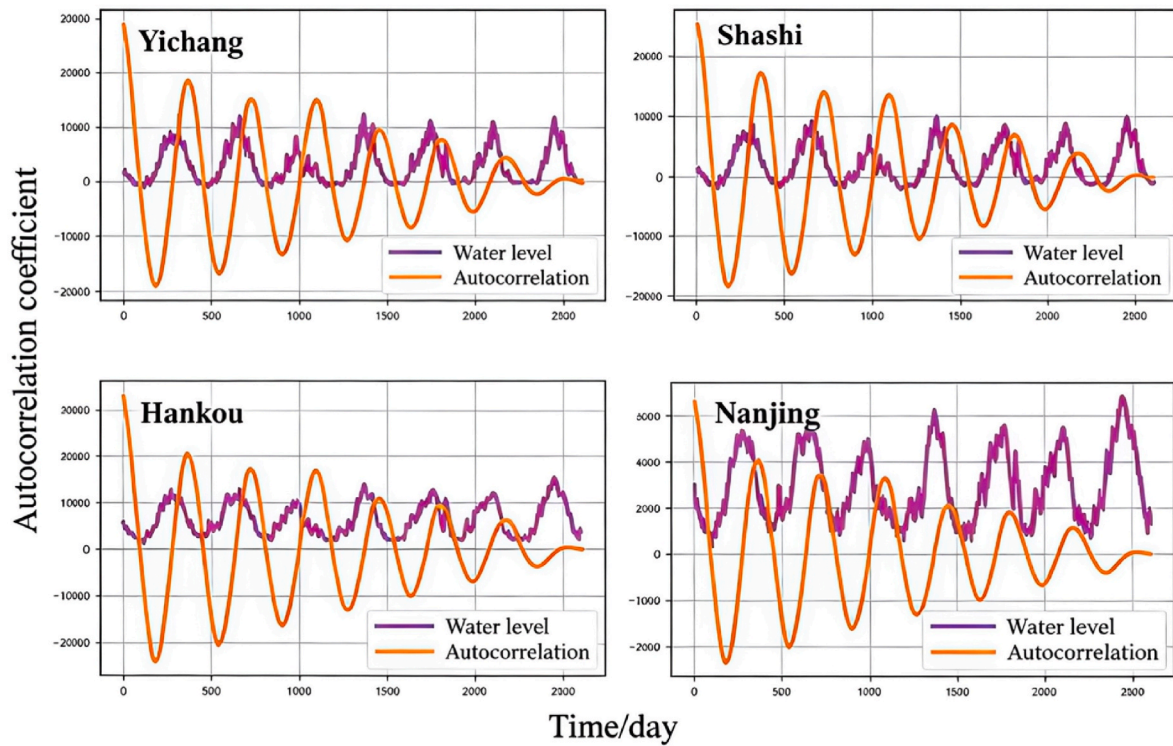


Fig. 3. Water level time series and autocorrelation coefficients at four stations. Purple curve: water level (m); orange curve: autocorrelation coefficient. The x-axis represents lag time (day).

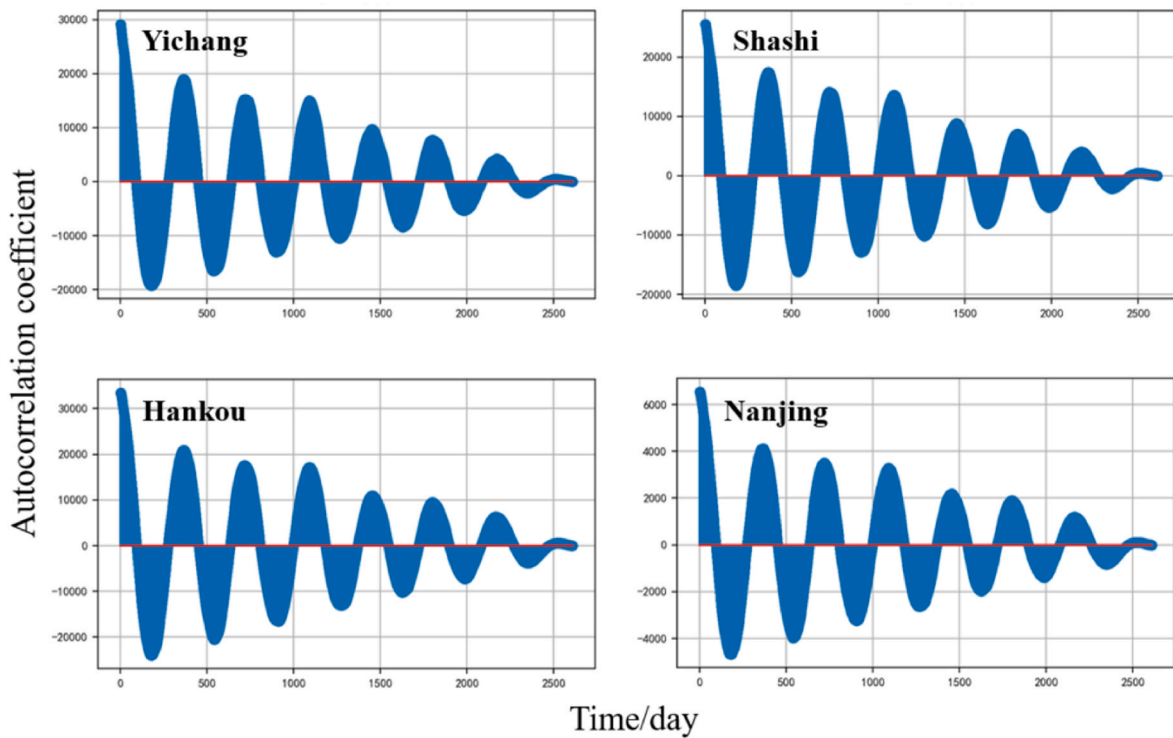


Fig. 4. Bar charts of autocorrelation coefficients at four stations. Bars above zero indicate positive autocorrelation, and bars below zero indicate negative autocorrelation. Dashed lines represent 95% confidence bounds.

b) Semi-annual negative autocorrelation ($k \approx 150\text{--}200$ days): The ACF crosses zero and reaches negative values, reflecting the hydrological contrast between the wet season (June–September) and the dry season (December–March).

c) Annual positive autocorrelation ($k \approx 365$ days): Positive autocorrelation returns, confirming a strong annual periodicity driven by the East Asian monsoon cycle.

By identifying the time lags at which the first three positive peaks emerge, we documented these in Table 2. The mean timings for the first three peaks are approximately 363.6 days, 716.8 days, and 1060.5 days, respectively. This pattern confirms that the dominant cyclical movement aligns closely with an annual timeframe, with minor inter-station variations reflecting local hydrological differences.

These findings guided the selection of the input window length for the prediction models. To capture the strong short-term autocorrelation while maintaining computational efficiency for one-day-ahead forecasting, we set the input window to 16 days. The sensitivity to this choice is discussed in Section 5.3.

3.3. Data normalization

To facilitate model training, we normalize the data to the range $[-1, 1]$ using min-max scaling:

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

where $x_{\max}$ and $x_{\min}$ are the maximum and minimum values in the training set, respectively, and x'_i is the normalized value corresponding to the original observation x_i . Critically, the normalization parameters ($x_{\max}$ and $x_{\min}$) are computed exclusively from the training set and applied identically to the validation and test sets to prevent information leakage. Predictions are inverse-transformed to the original scale before computing evaluation metrics. The normalization parameters are saved for consistent application during model testing.

3.4. Data splitting strategy

For time series prediction, temporal ordering must be strictly preserved to prevent future information leakage [31]. We adopt a chronological split of the total 2610 data points.

- a) Training set: 70% of the data (1826 days), representing the earliest portion of the time series
- b) Validation set: 10% of the data (256 days), immediately following the training period, used for hyperparameter tuning and early stopping
- c) Test set: 20% of the data (528 days), comprising the most recent portion, used exclusively for final evaluation

The test set encompasses approximately one full annual cycle (391 data points spanning from approximately December 5th to December 31st of the subsequent year), covering both dry and wet season variations. This division applies equally to both single and stacked models.

3.5. Prediction task formulation

Water level time series exhibit strong temporal autocorrelation (as demonstrated in Section 3.2), so the chronological order of observations must be strictly preserved throughout data preparation—no shuffling is applied at any stage.

The prediction task is formulated as one-day-ahead forecasting: given the observed daily water levels over the most recent 16 consecutive days, predict the water level on the following day. Formally:

Table 2

Time lags (days) of the first three positive autocorrelation peaks at four stations.

Station	First Peak	Second Peak	Third Peak
Yichang	359 d	739 d	1127 d
Shashi	346 d	694 d	937 d
Hankou	364 d	715 d	1089 d
Nanjing	366 d	719 d	1089 d
Mean	363.8 d	716.8 d	1060.5 d

$$\hat{x}_{t+1} = f(x_{t-15}, x_{t-14}, \dots, x_t) \quad (3)$$

where x_{t-k} denotes the observed water level k days before the current time step t , and $\hat{x}_{t+1}$ is the model's prediction for day $t + 1$. The input window length of 16 days was selected based on the autocorrelation analysis in Section 3.2, which shows strong short-term persistence over this timescale; sensitivity to this choice is discussed in Section 5.3.

Training samples are constructed using a rolling window approach: the 16-day input window is shifted forward by one day at a time across the full dataset, generating a new input–output pair at each step. This procedure maximizes the use of available training data while ensuring that each sample maintains the correct temporal ordering. Fig. 5 illustrates the complete data preparation workflow.

3.6. Model architecture

3.6.1. Overview of the RGL-stacking framework

The RGL-Stacking framework is organized as a two-level hierarchical architecture, as illustrated in Fig. 6.

Level 1 — Base Learners: Three recurrent neural networks with structurally distinct architectures—a vanilla RNN, a GRU, and an LSTM—each independently receive the same 16-day water level input sequence and produce a scalar prediction for the next day's water level. The three models are trained separately, each with its own optimizer, learning rate, and early stopping criterion, so that each model is free to specialize according to its architectural inductive bias (see Section 3.7 for a detailed discussion of their complementary roles).

Level 2 — Meta-Learner: The three scalar predictions generated by the base learners are concatenated into a 3-dimensional meta-feature vector. This vector is passed to a Meta-MLP—a compact multilayer perceptron with two hidden layers—which learns a nonlinear function that maps the three base predictions to a single final water level forecast. Crucially, the Meta-MLP is trained on out-of-fold predictions generated through a time-series-aware cross-validation procedure (Section 3.9), preventing information leakage from the base model training stage.

This two-level design allows the framework to capture a broader range of temporal patterns than any individual model, while the Meta-MLP adaptively learns which base learner to trust more under different hydrological conditions.

3.6.2. Base model 1: recurrent neural network (RNN)

The vanilla RNN computes hidden states through the recurrence:

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b_h) \quad (4)$$

where $h_t \in \mathbb{R}^d$ is the hidden state at time step t , $x_t \in \mathbb{R}$ is the input (water

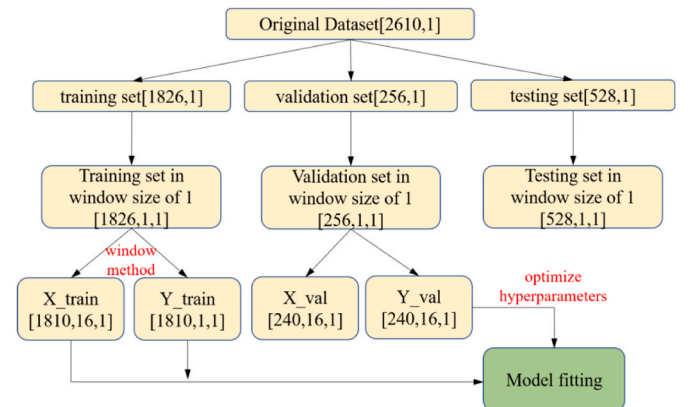


Fig. 5. Data preparation process for one-day-ahead prediction using a 16-day sliding window. The original data are split into training, validation, and test sets and then converted into input–output supervised learning samples.

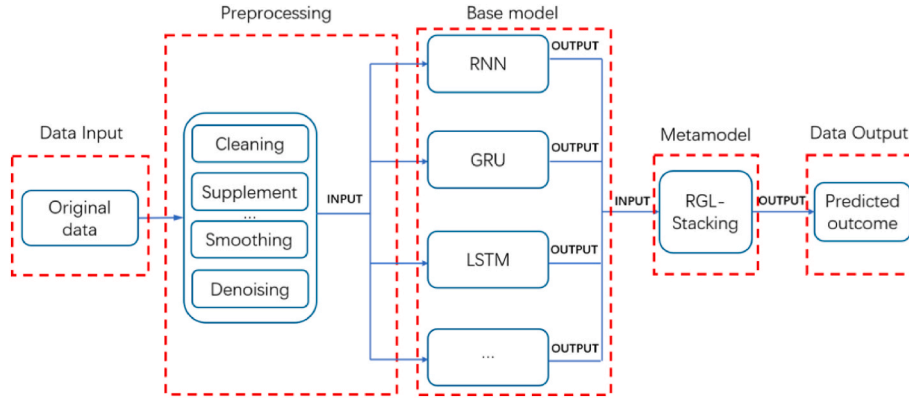


Fig. 6. Architecture of the proposed RGL-Stacking model. Three base learners (RNN, GRU, LSTM) generate predictions in parallel; predictions are concatenated as meta-features and fed into the Meta-MLP to produce the final forecast.

level) at time step t , $W_h \in \mathbb{R}^{d \times d}$ is the weight matrix for the hidden-to-hidden connection, $W_x \in \mathbb{R}^{d \times 1}$ is the weight matrix for the input-to-hidden connection, $b_h \in \mathbb{R}^d$ is the bias vector, σ is the activation function (tanh or ReLU), and d is the hidden state dimension. The final hidden state is mapped to the prediction through a fully connected output layer.

We design the RNN model with two hidden layers: the first layer is equipped with a substantial number of neurons to augment the model's representational capacity, and the second layer uses a reduced neuron count to facilitate the extraction of more refined abstract features. The RNN is most responsive to short-range temporal dynamics but is susceptible to the vanishing gradient problem over long sequences [32], which limits its ability to capture dependencies beyond the input window.

3.6.3. Base model 2: long short-term memory (LSTM)

The LSTM addresses the vanishing gradient problem through a memory cell C_t regulated by three gates [33]:

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (5)$$

Input gate:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (6)$$

Output gate:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (7)$$

Candidate cell state:

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (8)$$

Cell state update:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \quad (9)$$

Hidden state:

$$h_t = o_t \odot \tanh(C_t) \quad (10)$$

where $f_t \in \mathbb{R}^d$ is the forget gate activation that controls what information to discard from the previous cell state, $i_t \in \mathbb{R}^d$ is the input gate activation that controls what new information to store, $o_t \in \mathbb{R}^d$ is the output gate activation that controls what part of the cell state to output, $\tilde{C}_t$ is the candidate cell state containing new information to potentially add, C_t is the updated cell state, σ denotes the sigmoid function, $\odot$ denotes element-wise (Hadamard) multiplication, $[h_{t-1}, x_t]$ denotes concatenation of the previous hidden state and current input, and W_f, W_i, W_o, W_C and b_f, b_i, b_o, b_C are the learnable weight matrices and bias vectors for

each gate, respectively.

We construct a dual-layer LSTM: the first layer captures a broad spectrum of temporal features, while the second layer (with half the neurons) enables focused extraction of nuanced trends. A fully connected output layer generates the predicted water level. This gating mechanism enables LSTM to selectively retain or discard information over long time spans, making it well suited for capturing seasonal and inter-annual patterns in water level data.

3.6.4. Base model 3: Gated Recurrent Unit (GRU)

The GRU simplifies the LSTM architecture by merging the forget and input gates into a single update gate and eliminating the separate memory cell [34]:

Update gate:

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \quad (11)$$

Reset gate:

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \quad (12)$$

Candidate hidden state:

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h) \quad (13)$$

Hidden state update:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (14)$$

where $z_t \in \mathbb{R}^d$ is the update gate that determines how much of the previous hidden state to retain versus how much to update with new candidate information, $r_t \in \mathbb{R}^d$ is the reset gate that controls how much of the previous hidden state is used to compute the candidate, $\tilde{h}_t$ is the candidate hidden state, and W_z, W_r, W_h and b_z, b_r, b_h are the learnable weight matrices and bias vectors for each gate. With approximately 25% fewer parameters than LSTM (for the same hidden dimension), GRU offers competitive sequence modeling with improved computational efficiency [35].

3.6.5. Meta-model: Meta-MLP

The meta-model is a multi-layer perceptron (MLP) with three fully connected layers that utilizes the predictions from the base models to perform further learning, thereby enhancing prediction stability and robustness across different stations. The predictions from the three base models ($Y_{\text{predRNN}}, Y_{\text{predLSTM}}, Y_{\text{predGRU}}$) are concatenated to form the meta-feature input:

$$X_{\text{meta}} = [Y_{\text{predRNN}}, Y_{\text{predLSTM}}, Y_{\text{predGRU}}] \quad (15)$$

where $X_{\text{meta}} \in \mathbb{R}^3$ is the meta-feature vector for each time step.

The computations for each layer are:

$$h_1 = \sigma(W_1 X_{\text{meta}} + b_1) \quad (16)$$

$$h_2 = \sigma(W_2 h_1 + b_2) \quad (17)$$

$$Y_{\text{final}} = W_3 h_2 + b_3 \quad (18)$$

where $W_1 \in \mathbb{R}^{d_1 \times 3}$, $W_2 \in \mathbb{R}^{d_2 \times d_1}$, $W_3 \in \mathbb{R}^{1 \times d_2}$ are weight matrices, b_1 , b_2 , b_3 are bias vectors, d_1 and d_2 are the hidden dimensions of the first and second layers respectively, σ is the ReLU activation function, h_1 and h_2 are the hidden layer outputs, and Y_{final} is the final predicted water level. The hidden layer dimensions are progressively reduced to facilitate feature compression and consolidation.

3.7. Complementarity of base learners

The theoretical basis for stacking ensembles is that diverse base learners make different types of errors, and a meta-learner can exploit this diversity to produce a more accurate combined prediction [26]. The three recurrent architectures selected for RGL-Stacking differ fundamentally in their internal memory mechanisms, which leads to systematic differences in their forecasting behavior.

- a) RNN uses a simple recurrent update with no gating mechanism, making it highly responsive to recent observations. It is well suited for capturing short-range temporal dynamics—such as day-to-day water level fluctuations during the dry season—but suffers from the vanishing gradient problem [32], which limits its ability to retain information over periods longer than a few days.
- b) LSTM introduces a dedicated memory cell regulated by three learnable gates (input, forget, and output), enabling it to selectively retain or discard information over extended time spans. This makes LSTM particularly effective at modeling seasonal and inter-annual patterns in water level data. The trade-off is a higher parameter count (approximately 25% more than GRU for the same hidden dimension), which may increase the risk of overfitting when training data are limited.
- c) GRU consolidates the LSTM's three gates into two (update and reset), eliminating the separate cell state. This simplification reduces the parameter count by approximately 25% relative to LSTM while retaining competitive sequence modeling capability across multiple temporal scales, offering a favorable balance between expressiveness and parameter efficiency [35].

Because each architecture is sensitive to different temporal scales—short-range for RNN, long-range for LSTM, and intermediate for GRU—their prediction errors are systematically different rather than correlated. The Meta-MLP is trained to exploit this complementarity by learning which base learner's prediction is most reliable under each combination of input conditions. Empirical verification of this complementarity through residual correlation analysis is provided in Section 5.1.

3.8. Distinction from existing hybrid and ensemble models

To address the reviewer's concern regarding the novelty of RGL-Stacking relative to existing hybrid and ensemble deep learning

approaches, we provide an explicit and systematic comparison along five dimensions. Table 3 summarizes the structural distinctions.

- a) Architecture topology — parallel vs. serial: The most widely adopted hybrid models in hydrology, such as CNN-LSTM [20,21] and wavelet-LSTM [25], follow a serial (pipeline) topology in which a feature transformation module feeds sequentially into a sequence modeling module. This design introduces two critical vulnerabilities: (i) representation bottlenecks, where the downstream recurrent module can only access features filtered by the upstream extractor; and (ii) irreversible error propagation, where extraction errors in early stages cannot be corrected by later stages. By contrast, RGL-Stacking adopts a parallel topology: RNN, GRU, and LSTM independently process the identical raw input sequence and generate predictions without any inter-model dependency. Each base learner is free to develop its own internal representation, and no prediction error from one model contaminates another. The Meta-MLP then synthesizes all three outputs through a learned nonlinear combination.
- b) Training regime — independent vs. end-to-end joint: Serial hybrid models and attention-augmented recurrent networks (e.g., Wang et al. [22]) are typically trained end-to-end with shared gradients flowing through the entire architecture. While end-to-end training can optimize global loss, it forces intermediate representations to satisfy multiple objectives simultaneously, potentially compromising the specialized capability of individual components. In RGL-Stacking, each base model (RNN, GRU, LSTM) is trained independently with its own optimizer, loss function, and early stopping criterion. This preserves each model's unique inductive bias: the RNN retains its sensitivity to short-range dynamics, the LSTM maintains its long-range memory specialization, and the GRU preserves its parameter-efficient intermediate representation. The Meta-MLP is subsequently trained separately on out-of-fold predictions, ensuring a clean separation of optimization objectives.
- c) Ensemble combination mechanism — nonlinear learned vs. fixed: Conventional ensemble methods in hydrology predominantly use simple averaging or linear weighting (including ridge regression stacking) to combine base model outputs [17,28]. These approaches impose a rigid linear combination structure that cannot adapt to the context-dependent complementarity of base learners—for instance, the RNN may be more reliable during short-term dry-season fluctuations while the LSTM provides superior accuracy during flood-season peaks. The Meta-MLP in RGL-Stacking learns a fully nonlinear combination function with hidden dimensions of 64 and 32 neurons, enabling it to dynamically weight base learner contributions based on their predicted values. This nonlinear meta-learning is empirically shown to outperform linear alternatives through the ablation study in Section 5.2.
- d) Architectural diversity — homogeneous vs. heterogeneous base learners: Homogeneous ensembles such as bagging multiple LSTM instances improve variance reduction but are limited in bias correction because all members share the same functional form and inductive bias [23,24]. Even the stacking approach in Granata et al. [28], which combines traditional machine learning algorithms (e.g., Random Forest, SVM) with Bi-LSTM, does not systematically

Table 3

Hyperparameter settings for all models.

Dimension	Serial Hybrid (e.g., CNN-LSTM)	Homogeneous Ensemble	Heterogeneous Linear Stacking	RGL-Stacking (Proposed)
Topology	Serial pipeline	Parallel (same arch.)	Parallel (mixed)	Parallel (structurally diverse)
Training	End-to-end joint	Independent	Independent	Fully independent
Combination	Fixed (concatenation)	Averaging/voting	Linear weighting	Nonlinear Meta-MLP
Base learner diversity	Low (fixed roles)	None	Medium	High (RNN/GRU/LSTM)
Leakage prevention	N/A	N/A	Rarely addressed	Time-series blocked CV
Error propagation	Yes	No	No	No

leverage the complementary temporal modeling capabilities across the RNN family. RGL-Stacking deliberately selects three architecturally distinct recurrent models: the vanilla RNN (no gating, purely short-range dynamics), GRU (dual-gate design, intermediate memory), and LSTM (triple-gate with explicit cell state, long-range dependencies). These differences in gating mechanisms produce systematically different error patterns, as quantified by the inter-model residual correlations in Section 5.1 (mean Pearson $r = 0.53\text{--}0.71$, well below the redundancy threshold of 0.90 [36]).

- e) Information leakage prevention — time-series-aware cross-validation: A frequently overlooked issue in stacking ensembles for time series is information leakage: if base model predictions on training data are generated using models already trained on that data, the meta-learner learns to exploit overfitting artifacts rather than genuine complementarity. Most existing hydrological ensemble studies do not explicitly address this issue. RGL-Stacking employs a time-series-aware blocked cross-validation procedure with five chronological folds and a mandatory 16-day gap between training and prediction blocks (equal to the input window length), ensuring that the meta-feature matrix used to train the Meta-MLP is entirely free of information leakage (Section 3.9).

3.9. Training strategy and information leakage prevention

A well-known but frequently overlooked issue in stacking ensembles is information leakage between the base model training stage and the meta-learner training stage. If the meta-learner is trained on predictions that the base models generated from the same data they were trained on, those predictions will reflect the base models' training errors rather than their generalization errors. The meta-learner will then learn to correct training-specific artifacts, yielding overoptimistic performance estimates that do not generalize to unseen data [26].

To prevent this, we adopt a time-series-aware blocked cross-validation procedure specifically designed for sequential data. The key principle is that, for each fold, the base models must generate predictions on data they have never seen during training. The procedure consists of four steps.

- Block partitioning:** The 1826-day training period is divided into five non-overlapping temporal blocks of approximately equal length, in strict chronological order: Block 1 (days 1–365), Block 2 (days 366–730), Block 3 (days 731–1095), Block 4 (days 1096–1461), and Block 5 (days 1462–1826).
- Out-of-fold prediction:** For each fold $k \in \{2, 3, 4, 5\}$, the base models are trained on blocks 1 through $k-1$ and then used to generate predictions for block k . A 16-day gap is enforced between the end of the training blocks and the start of the prediction block; this gap equals the input window length and ensures that no sliding window used during prediction overlaps with any training observation. For fold $k = 1$, which has no preceding training blocks, a warm-start strategy is used: the base models are trained on the first 80% of Block 1 and predict the remaining 20%.
- Meta-feature construction:** The out-of-fold predictions from all five folds are concatenated in chronological order to form the meta-feature matrix $X_{\text{meta}} \in \mathbb{R}^{N \times 3}$, where N is the total number of training samples. Because each prediction was generated by a model that had no access to the corresponding observation during training, the meta-feature matrix is free of information leakage.
- Final base model retraining:** After the meta-feature matrix has been constructed, each base model is retrained from scratch on the entire training set. These retrained models, together with the Meta-MLP trained on X_{meta} , form the final deployed RGL-Stacking system.

The 16-day gap between training and prediction blocks is a deliberate design choice: it matches the input window length, guaranteeing that the first prediction window in each fold does not overlap with any

training observation. This design balances leakage prevention with training data utilization efficiency.

3.10. Training details

In both the base models and the meta-model, we employ the Rectified Linear Unit (ReLU) as the activation function for hidden layers, driven by its computational efficiency and its capacity to mitigate the vanishing gradient problem [37]. The Adam optimizer [38] is used for all models, chosen for its ability to dynamically adjust the learning rate throughout training, thereby facilitating faster convergence.

Throughout training, we implement an Early Stopping mechanism to prevent overfitting: training ceases if the mean squared error (MSE) on the validation set fails to show improvement over 5 consecutive epochs. The early stopping criterion monitors the validation MSE, and the model weights from the epoch with the lowest validation MSE are retained for final evaluation.

To determine effective hyperparameter configurations, we employ Grid Search on the validation set, exploring combinations of learning rates ($\{0.001, 0.005, 0.01\}$), hidden layer dimensions ($\{32, 64, 128\}$), and batch sizes ($\{16, 32, 64\}$). Table 4 presents the final hyperparameter settings determined through this process.

In summary, for each base model M_i (where $i \in \{\text{RNN, LSTM, GRU}\}$), the training process involves:

$$Y_{\text{pred}_i} = M_i(X_{\text{train}}) \quad (19)$$

Combining the prediction results of all base models into the meta-feature matrix:

$$X_{\text{meta}} = [Y_{\text{predRNN}}, Y_{\text{predLSTM}}, Y_{\text{predGRU}}] \quad (20)$$

Finally, the meta-model M_{meta} is trained using X_{meta} as input and Y_{train} as the target:

$$Y_{\text{final}} = M_{\text{meta}}(X_{\text{meta}}) \quad (21)$$

3.11. Evaluation metrics

To comprehensively quantify model performance, we employ five evaluation metrics:

Nash–Sutcliffe Efficiency (NSE) [39]:

$$\text{NSE} = 1 - \frac{\sum_{t=1}^n (x_t - \hat{x}_t)^2}{\sum_{t=1}^n (x_t - \bar{x})^2} \quad (22)$$

where x_t is the observed water level at time t , $\hat{x}_t$ is the predicted water level, $\bar{x}$ is the mean of all observed values, and n is the total number of predictions. NSE ranges from $-\infty$ to 1, where $\text{NSE} = 1$ indicates perfect

Table 4
Hyperparameter settings for all models.

Hyperparameter	RNN	GRU	LSTM	Meta-MLP
Hidden dimensions (Layer 1/Layer 2)	128/ 64	128/ 64	128/ 64	64/32
Number of recurrent layers	2	2	2	—
Number of fully connected layers	1	1	1	3
Activation function	ReLU	ReLU	ReLU	ReLU (hidden); Linear (output)
Optimizer	Adam	Adam	Adam	Adam
Learning rate	0.001	0.001	0.001	0.005
Batch size	32	32	32	64
Max epochs	200	200	200	100
Early stopping patience	5	5	5	5
Loss function	MSE	MSE	MSE	MSE
Input window (days)	16	16	16	3 (meta-features)

prediction, $NSE = 0$ indicates the model performs no better than using the observed mean, and $NSE < 0$ indicates the model is worse than the mean. NSE is the standard benchmark metric in hydrology [40] and directly measures whether the model captures variability beyond the climatological mean.

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{t=1}^n (x_t - \hat{x}_t)^2}{\sum_{t=1}^n (x_t - \bar{x})^2} \quad (23)$$

Note that in this formulation, R^2 is numerically identical to NSE . We retain both notations for consistency with the hydrological (NSE) and machine learning (R^2) communities.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{t=1}^n |x_t - \hat{x}_t| \quad (24)$$

where $|\cdot|$ denotes absolute value. MAE provides an intuitive measure of average prediction error in the original unit (meters).

Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{t=1}^n (x_t - \hat{x}_t)^2 \quad (25)$$

MSE penalizes larger errors more heavily than MAE due to the squaring operation.

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (x_t - \hat{x}_t)^2} \quad (26)$$

RMSE shares the same units as the original data and provides a measure of typical error magnitude.

4. Results

4.1. Individual model performance

4.1.1. RNN model results

We initially utilized the RNN model to conduct predictions at the four stations. Fig. 7 illustrates the results. The test set encompasses 391 data points, spanning approximately one full year and covering both dry and wet season variations. In the figure, the x-axis denotes time in days,

while the y-axis measures the water level in meters. Actual water levels are traced by a red curve, whereas predicted levels are shown in blue (see Fig. 8).

The RNN model capably mirrors the fluctuating water levels, demonstrating notable predictive accuracy. The MAE values for the four stations are 0.287, 0.185, 0.060, and 0.119 m, respectively. The MAE at Hankou station falls below 0.1 m, suggesting relatively stable water level changes at this location. However, Yichang, Shashi, and Nanjing exhibit higher MAEs, particularly during the wet season (approximately days 175 to 275), where predictions diverge markedly from observations.

4.1.2. GRU model results

The GRU model not only preserves commendable prediction accuracy at Hankou but also notably enhances accuracy for Yichang, Shashi, and Nanjing, reducing their MAEs to 0.140, 0.102, and 0.061 m, respectively. However, the GRU model reveals a pronounced deviation near the peak water levels at Nanjing station (around days 200–230), underscoring some limitations.

4.1.3. LSTM model results

The LSTM model showcases performance similar to that of the GRU, registering MAEs of 0.1382, 0.1130, 0.1031, and 0.0553 m for the four stations. This reflects a more uniform improvement in predictive accuracy across various periods and conditions, highlighting the LSTM's balanced capability in forecasting water level changes.

4.2. RGL-stacking model performance

We applied the RGL-Stacking model for predictive analysis at the four stations, with findings depicted in Fig. 10 (see Fig. 9). The results underscore the stacking model's superiority over individual models in both accuracy and reliability. The MAE values—0.0866, 0.0683, 0.0595, and 0.0563 m—all fall below 0.1 m, signifying a marked improvement.

4.3. Comprehensive performance comparison

Table 5 compiles the complete error metrics for all models at all four stations, including MAE, MSE, RMSE, R^2 , and NSE .

Several key observations emerge from Table 5.

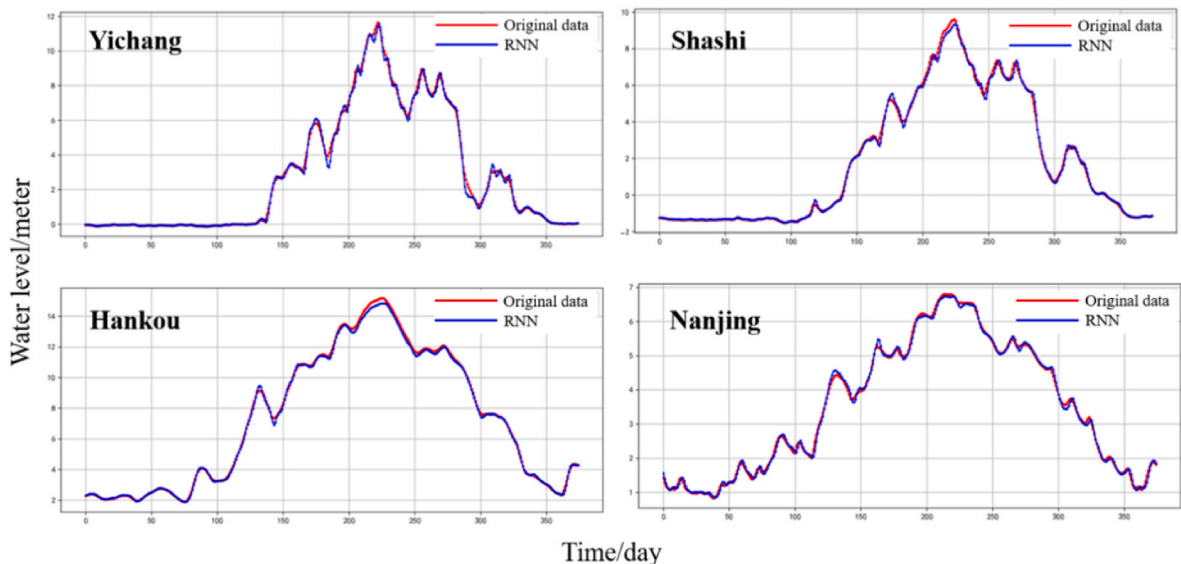


Fig. 7. Water level predictions by the RNN model at four stations. Red solid line: observed; blue solid line: predicted.

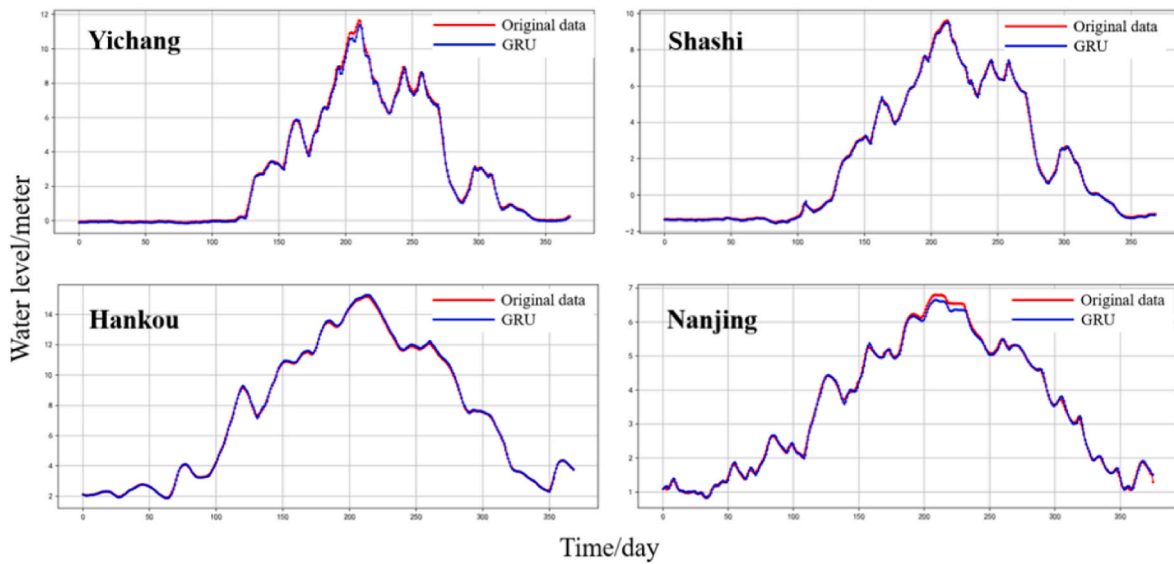


Fig. 8. Water level predictions by the GRU model at four stations. Red solid line: observed; blue solid line: predicted.

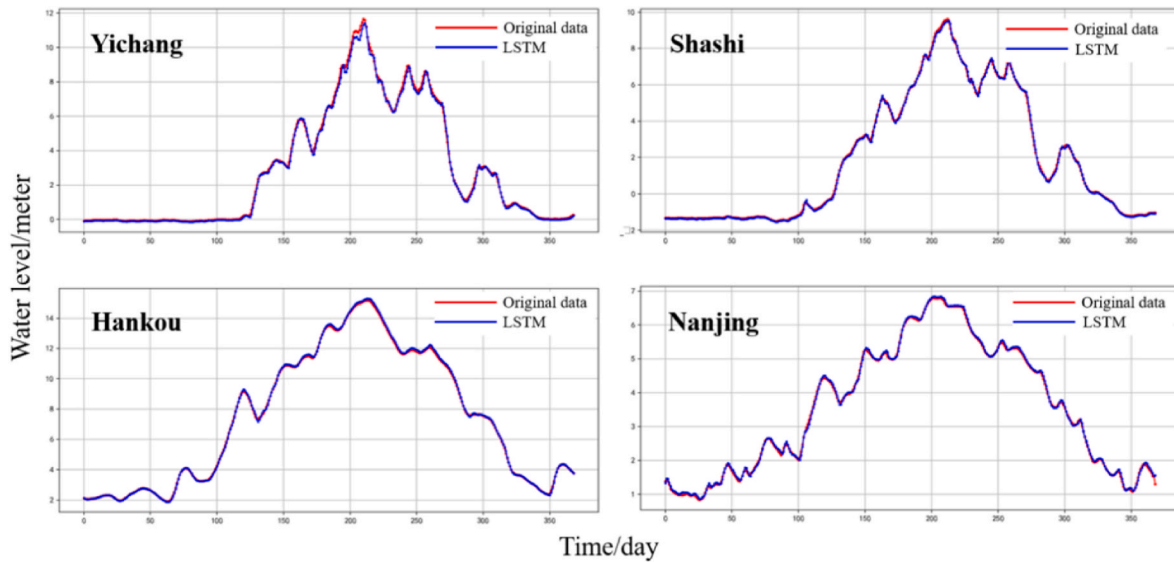


Fig. 9. Water level predictions by the LSTM model at four stations. Red solid line: observed; blue solid line: predicted.

- a) RGL-Stacking achieves the highest NSE at all four stations (0.998, 0.999, 0.999, and 0.998), demonstrating consistently superior accuracy. All NSE values exceed 0.99, indicating excellent predictive capability.
- b) The SVM model exhibits the lowest accuracy, with MAE values substantially higher than the deep learning models at all stations, confirming the advantage of recurrent architectures for temporal pattern recognition in water level data.
- c) Individual base models show station-dependent performance variation: the RNN model yields the lowest MAE at Hankou (0.060 m), the GRU model performs best at Shashi (0.103 m), and the LSTM model is most accurate at Yichang (0.133 m) and Nanjing (0.055 m). This variability across stations motivates the need for an ensemble approach.
- d) The Transformer (TRM) model performs slightly better than the stacking model in MAE, MSE, RMSE, and R^2 at some stations, but falls short of the stacking model in the most critical NSE metric. This is because the TRM may achieve lower absolute errors on average

while producing larger errors during critical periods that disproportionately affect the NSE denominator.

- e) RGL-Stacking achieves consistent performance across all stations, with MAE values all below 0.1 m, demonstrating the ensemble's ability to adapt to diverse hydrological conditions without manual station-specific tuning.

4.4. Dry season evaluation

The hydrological management system for the Yangtze River Navigation Channel faces increased demand for high-precision water level predictions during the dry season. Figs. 11–14 present detailed comparisons of prediction results from December 1st to April 1st of the subsequent year (approximately 120 days) across the four models at each station.

The comparative analysis reveals that the RGL-Stacking model consistently outperforms the three individual models in predicting water levels during the dry season across nearly all stations. Such enhanced accuracy is crucial for effective hydrological management, particularly

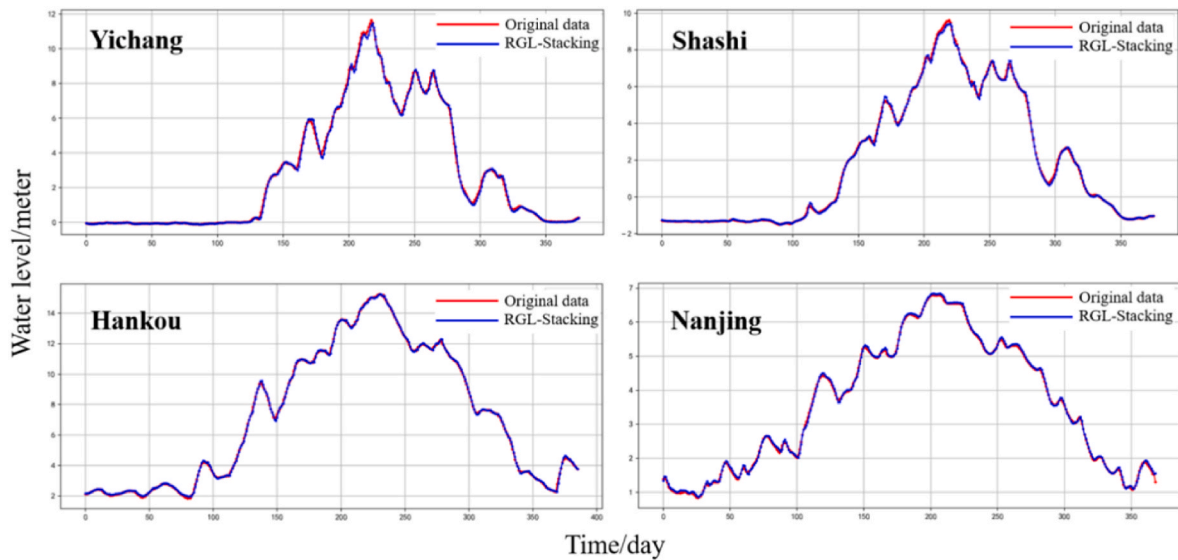


Fig. 10. Water level predictions by the RGL-Stacking model at four stations. Red solid line: observed; blue solid line: predicted.

Table 5

Performance metrics of different models at four hydrological stations. Bold values indicate the best performance.

Model	Yichang					Shashi				
	MAE	MSE	RMSE	R ²	NSE	MAE	MSE	RMSE	R ²	NSE
RNN	0.287	0.197	0.444	0.983	0.992	0.185	0.216	0.465	0.981	0.996
GRU	0.140	0.064	0.252	0.994	0.998	0.103	0.032	0.178	0.997	0.999
LSTM	0.133	0.038	0.199	0.996	0.997	0.113	0.024	0.154	0.998	0.996
TRM	0.045	0.002	0.050	0.991	0.989	0.038	0.002	0.042	0.995	0.995
SVM	0.402	0.196	0.443	0.983	0.983	0.252	0.091	0.302	0.992	0.992
RGL-Sta	0.087	0.020	0.142	0.998	0.998	0.068	0.011	0.102	0.999	0.999
Model	Hankou					Nanjing				
	MAE	MSE	RMSE	R ²	NSE	MAE	MSE	RMSE	R ²	NSE
RNN	0.060	0.006	0.079	0.999	0.999	0.119	0.029	0.171	0.991	0.996
GRU	0.069	0.009	0.096	0.999	0.999	0.062	0.008	0.087	0.998	0.999
LSTM	0.103	0.020	0.141	0.998	0.997	0.055	0.005	0.071	0.998	0.998
TRM	0.042	0.003	0.051	0.994	0.993	0.036	0.002	0.043	0.995	0.994
SVM	0.346	0.154	0.392	0.991	0.992	0.167	0.038	0.194	0.988	0.988
RGL-Sta	0.060	0.007	0.055	0.999	0.999	0.056	0.006	0.076	0.998	0.998

in optimizing navigation safety and water resource allocation during low-water periods.

4.5. Flood season evaluation

To assess model performance under the markedly different conditions of the flood season, characterized by rapid water level increases, we focus on days 170–220 (approximately mid-June to early August), a period typically indicative of the flood season. Figs. 15–18 present the comparisons.

The analysis reveals that predicting water levels during the flood season presents a more formidable challenge compared to the dry season. The individual models demonstrate a pronounced delay in accurately tracking swift water level changes. Despite these challenges, the RGL-Stacking model consistently offers better prediction accuracy and stability across different stations even during the flood season, underscoring its capability to adapt to rapidly changing conditions.

4.6. Error metric comparison across models

Figs. 19 and 20 present bar charts comparing the MAE and RMSE metrics across the four models at each station.

Across all four stations, the RGL-Stacking model consistently

achieves the best or near-best prediction accuracy. The comparison between individual models and the RGL-Stacking model demonstrates a clear advantage of the ensemble in terms of robustness and accuracy throughout the year and across different stations.

5. Discussion

5.1. Base model complementarity analysis

To verify that the three base models provide complementary rather than redundant information, we computed the Pearson correlation coefficients among their prediction residuals ($e_t = x_t - \hat{x}_t$) on the test set. Table 6 presents the results for each station.

The residual correlations are moderate (0.49–0.74) across all stations. Notably, RNN and LSTM residuals show the lowest correlation (mean $r = 0.53$), confirming that these two architecturally most distinct models make substantially different types of errors: the RNN tends to produce larger errors during flood season peaks (where long-range memory is needed), while the LSTM tends to show slightly higher errors during short-term fluctuations in the dry season. GRU shows intermediate correlation with both, consistent with its intermediate architectural design. These correlations are well below the threshold of 0.90 that would indicate redundancy in ensemble members [36],

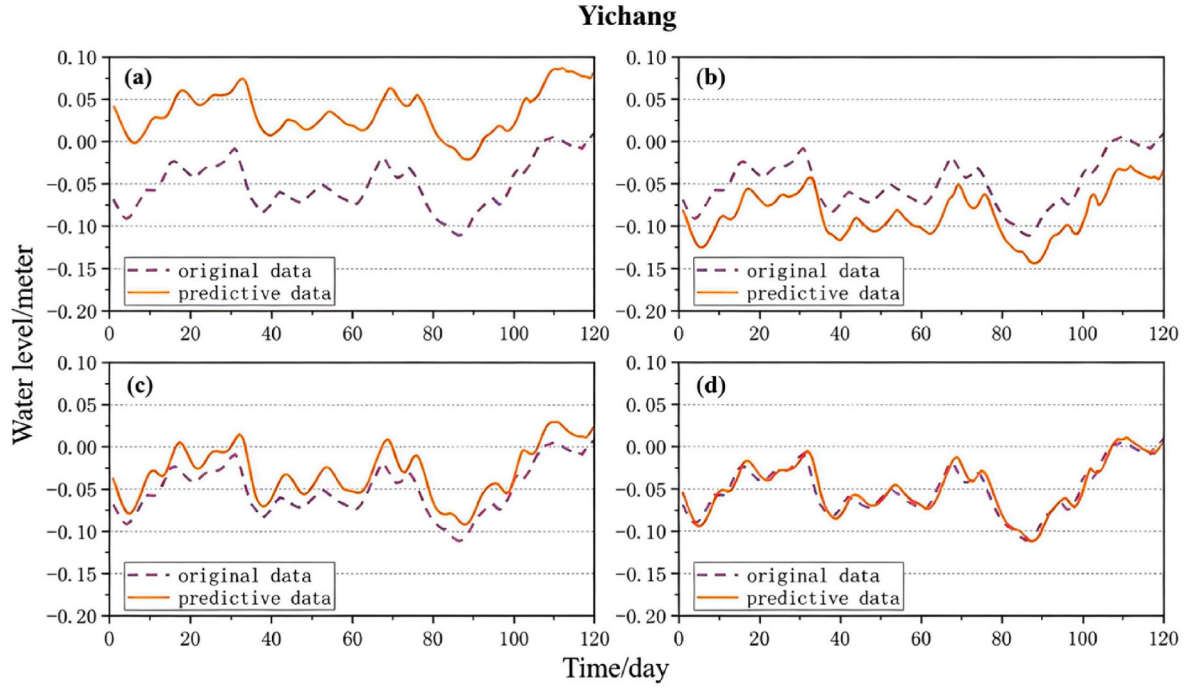


Fig. 11. Water level predictions during dry season at Yichang station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

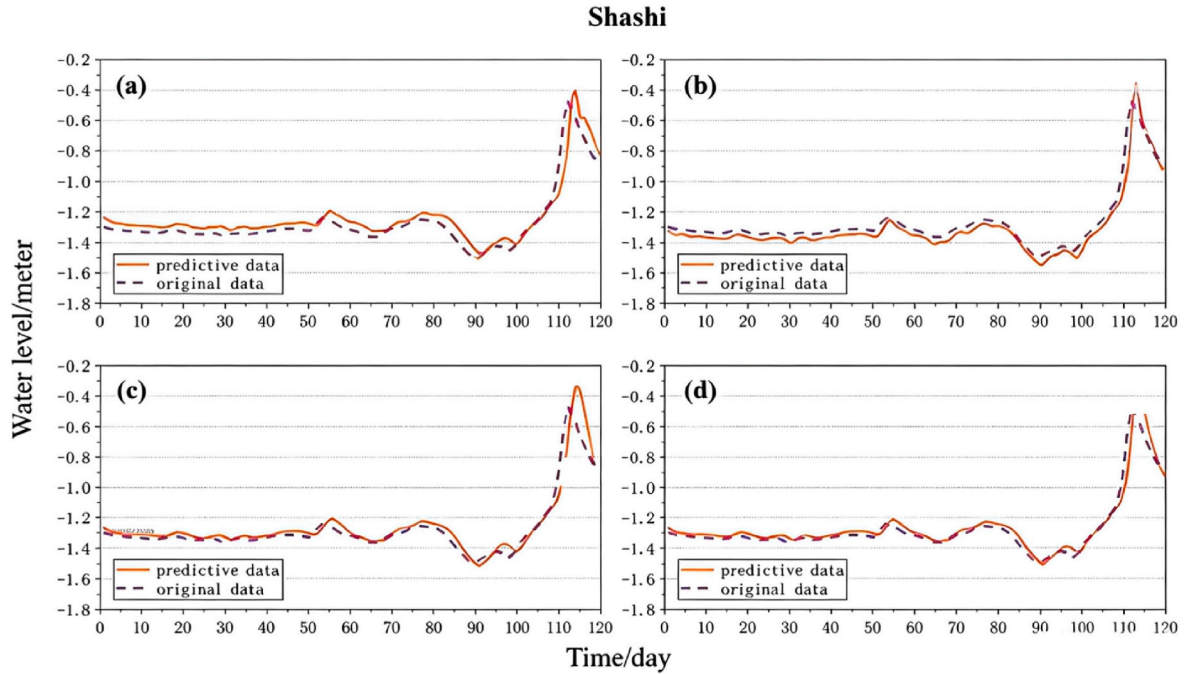


Fig. 12. Water level predictions during dry season at Shashi station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

confirming sufficient diversity for effective stacking.

5.2. Ablation study

To confirm that each base model contributes positively to the ensemble, we conducted ablation experiments comparing different subsets of base models, all using the same Meta-MLP architecture. Table 7 presents the NSE results at the four stations.

The full three-model ensemble achieves the highest or equal-highest NSE at all stations. While the differences are small (due to the already very high baseline NSE), the consistency of the full ensemble's advantage across all stations confirms the positive contribution of each base model. The RNN + LSTM combination (the most architecturally diverse pair) generally matches or approaches the full ensemble more closely than other pairs, supporting the importance of architectural diversity.

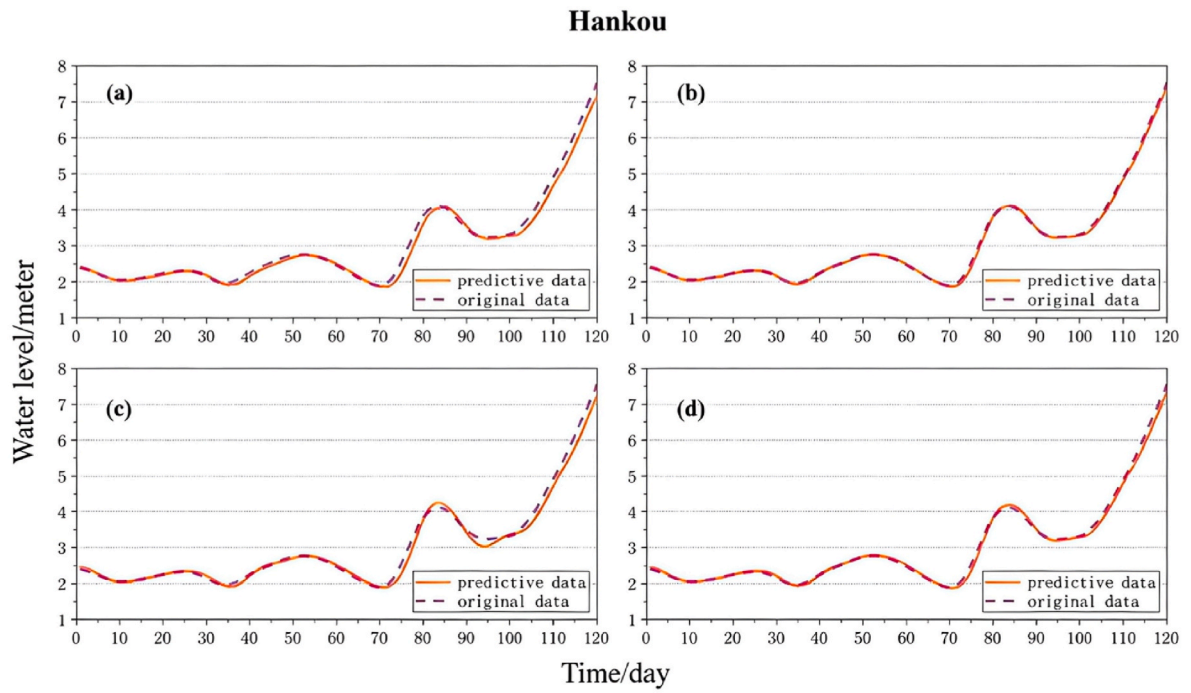


Fig. 13. Water level predictions during dry season at Hankou station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

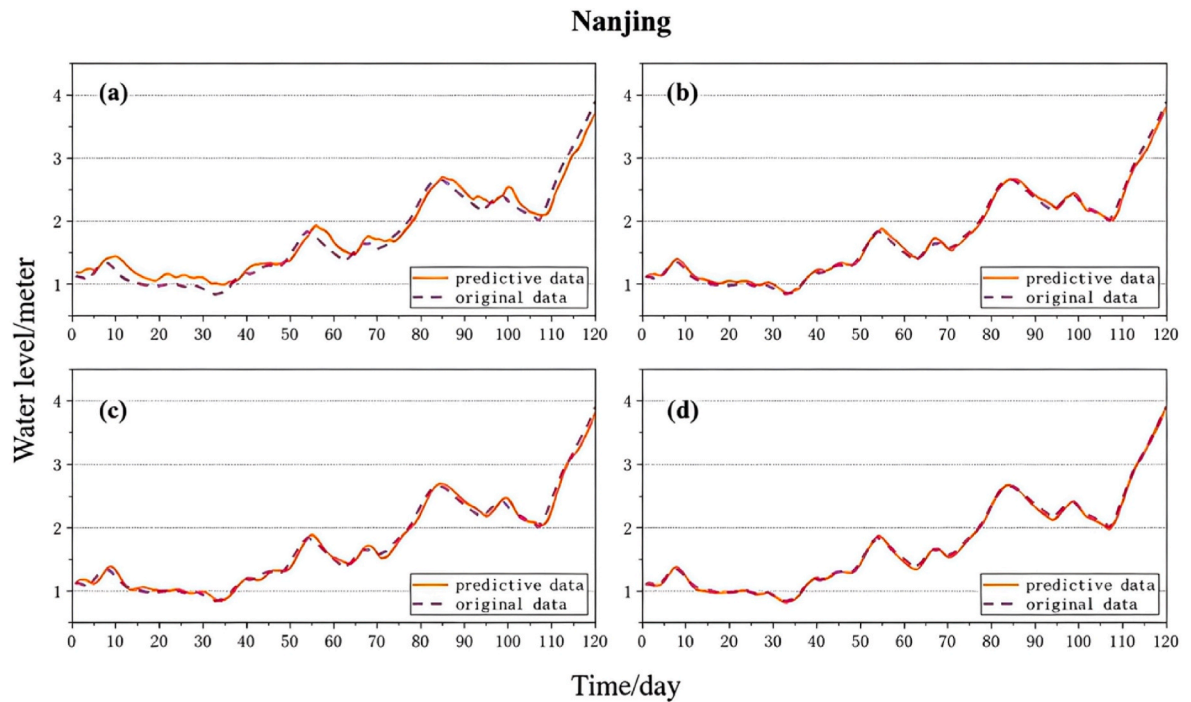


Fig. 14. Water level predictions during dry season at Nanjing station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

5.3. Input window length discussion

The current input window length of 16 days was selected based on preliminary experiments showing that this length captures sufficient short-term temporal dependencies for one-day-ahead prediction while maintaining computational efficiency. Although the autocorrelation analysis (Section 3.2) reveals dominant annual periodicity at

approximately 365 days, the one-day-ahead prediction task primarily relies on short-term persistence patterns, for which a 16-day window captures the most relevant recent dynamics.

Longer windows (e.g., 30, 60, or 365 days) could potentially capture additional seasonal context but would substantially increase computational cost and may introduce noise from less relevant historical information for the short-term prediction task. A systematic sensitivity

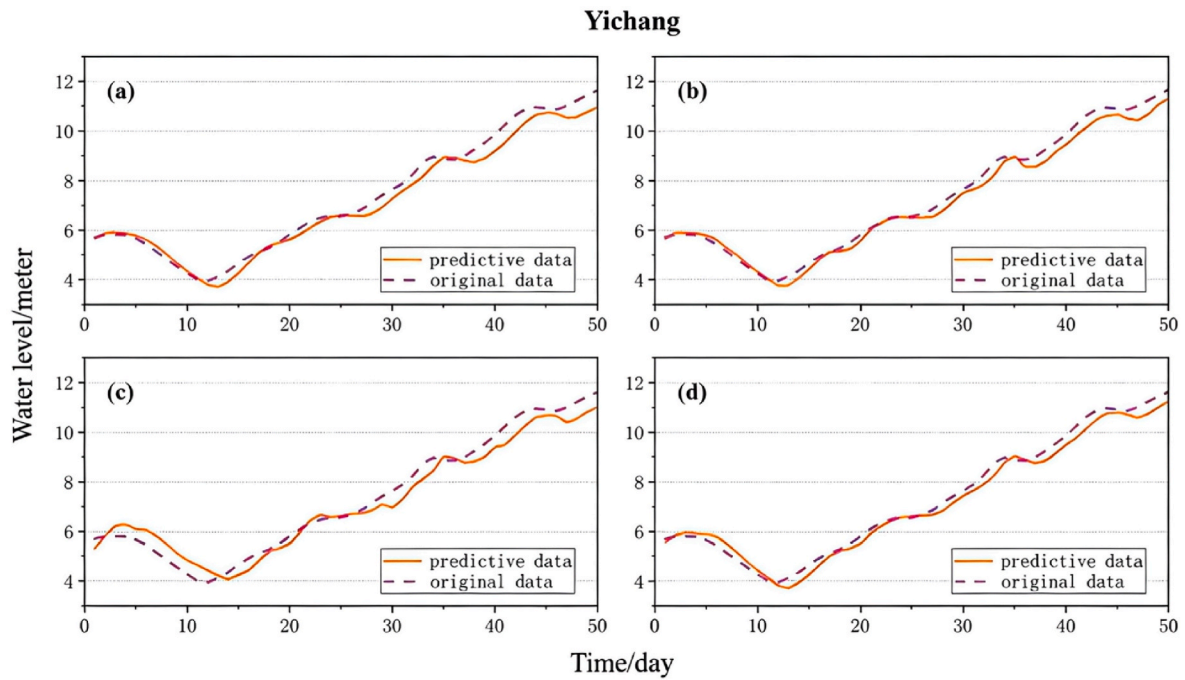


Fig. 15. Water level predictions during flood season at Yichang station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

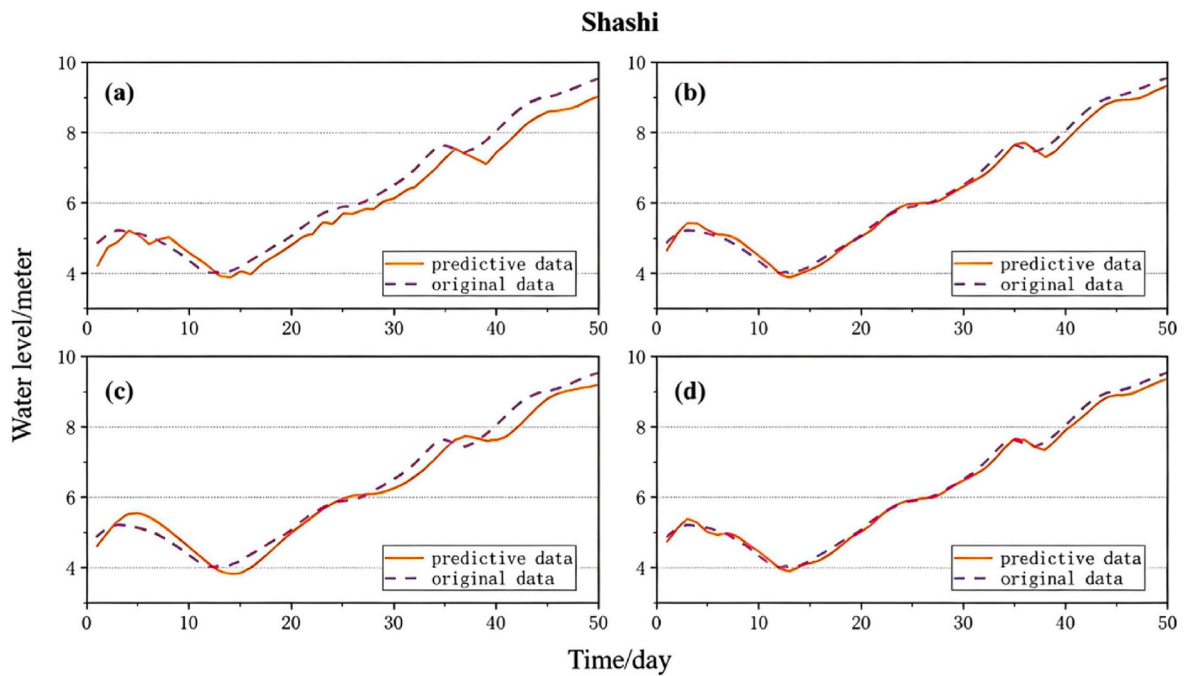


Fig. 16. Water level predictions during flood season at Shashi station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

analysis of window length is left as future work, particularly for extending the model to multi-day-ahead prediction where longer windows may provide greater benefit.

5.4. Computational cost analysis and real-time deployment feasibility

A practical concern for any operational hydrological forecasting system is whether the computational overhead of an ensemble model remains acceptable for real-time or near-real-time deployment. We

address this concern explicitly through three aspects: training cost, inference latency, and hardware requirements.

Training cost: Table 8 compares the computational cost of all methods, measured on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB), Intel i9-12900 K CPU, and 64 GB RAM. The sequential training time of RGL-Stacking is 17.3 min (sum of three base models plus 1.2 min for Meta-MLP training). Since the three base models are architecturally independent, they can be trained in parallel on multi-GPU or multi-process hardware, reducing the effective wall-clock

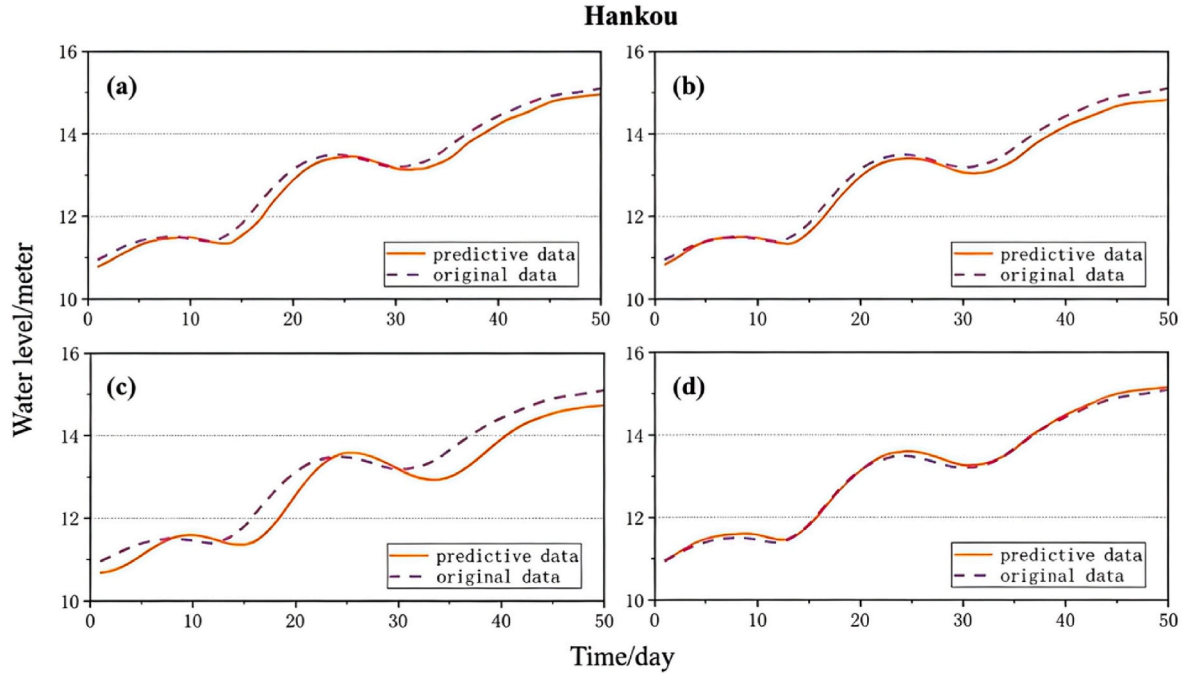


Fig. 17. Water level predictions during flood season at Hankou station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

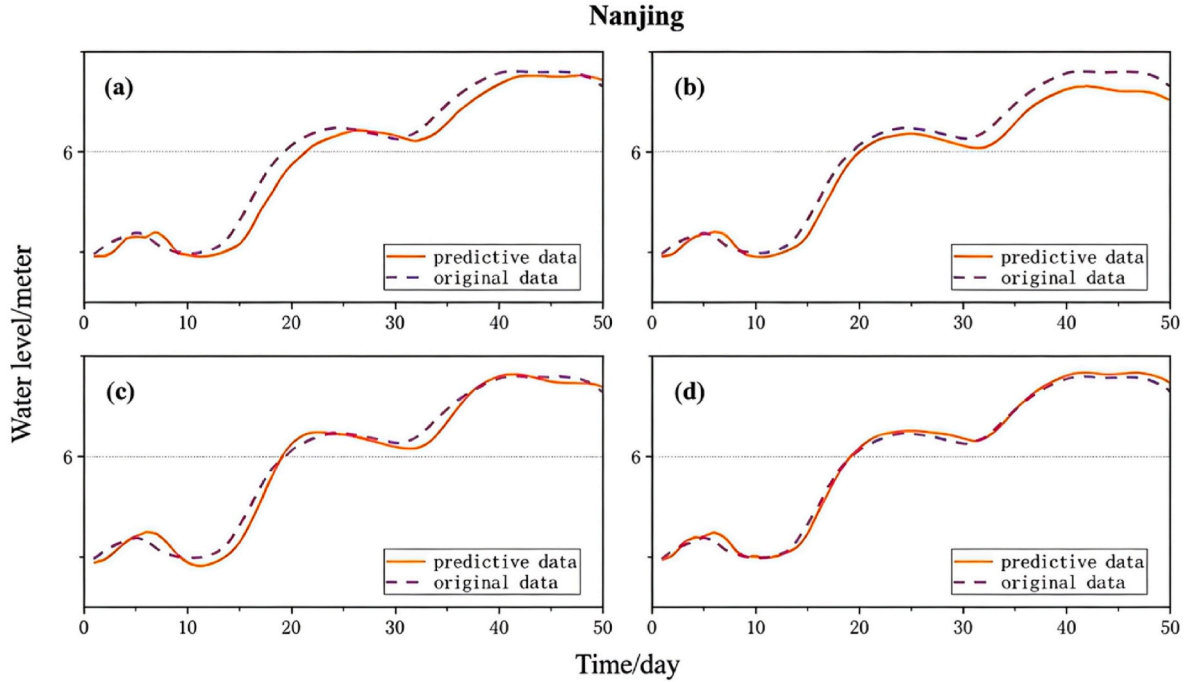


Fig. 18. Water level predictions during flood season at Nanjing station by different models: (a) RNN, (b) GRU, (c) LSTM, (d) RGL-Stacking. Dashed line: observed; solid line: predicted.

training time to approximately 8.5 min—comparable to training a single LSTM (6.7 min) plus a lightweight meta-learner. This represents approximately $2.1 \times$ the training time of a single LSTM when parallelized ($3.0 \times$ when sequential), which is considerably lower than the Transformer model (28.4 min, approximately $4.2 \times$ LSTM). Importantly, model training is an offline, periodic operation (conducted quarterly or after major hydrological regime changes), so even the sequential training time of 17.3 min is fully acceptable for operational systems.

Inference latency: For real-time river monitoring and flood early warning systems, the critical metric is inference latency rather than training time. As shown in Table 8, RGL-Stacking requires approximately 0.05 s per station for inference—only marginally higher than a single LSTM (0.02 s) and substantially lower than the Transformer (0.08 s). For a network of four stations, the total inference time is approximately 0.2 s, which is orders of magnitude faster than the operational update cycle (typically hourly or daily for major Chinese hydrological

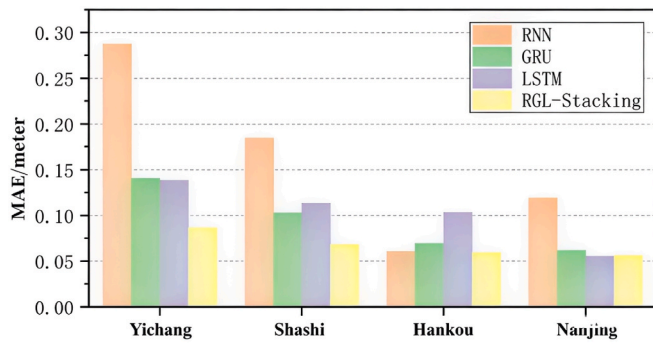


Fig. 19. MAE (m) of different models at four hydrological stations.

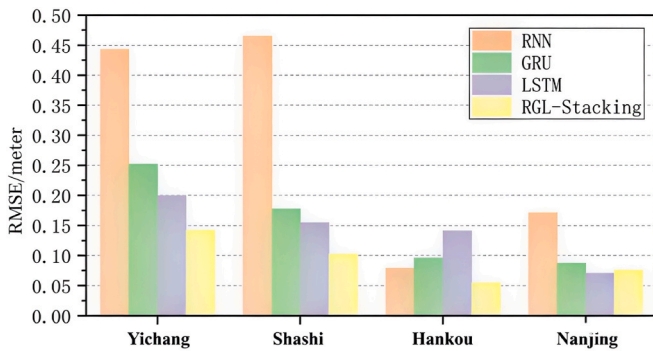


Fig. 20. RMSE (m) of different models at four hydrological stations.

Table 6

Pearson correlation coefficients of prediction residuals among base models on the test set.

Station	RNN-GRU	RNN-LSTM	GRU-LSTM
Yichang	0.68	0.54	0.72
Shashi	0.65	0.51	0.69
Hankou	0.71	0.58	0.74
Nanjing	0.63	0.49	0.67
Mean	0.67	0.53	0.71

Table 7

NSE values of different base model combinations in the ablation study.

Base Model Combination	Yichang	Shashi	Hankou	Nanjing
RNN alone	0.992	0.996	0.999	0.996
GRU alone	0.998	0.999	0.999	0.999
LSTM alone	0.997	0.996	0.997	0.998
RNN + GRU	0.998	0.999	0.999	0.998
RNN + LSTM	0.998	0.999	0.999	0.998
GRU + LSTM	0.997	0.998	0.999	0.998
RNN + GRU + LSTM	0.998	0.999	0.999	0.998

Table 8

Computational cost comparison of different models at Hankou station.

Model	Approximate Parameters	Training Time (min)	Inference Time (s)	Epochs to Convergence
RNN	49,793	3.8	0.01	78
GRU	116,225	5.6	0.02	85
LSTM	148,737	6.7	0.02	92
SVM	—	1.2	0.01	—
TRM	524,289	28.4	0.08	124
RGL-Stacking	318,850 ^a	17.3 ^b	0.05	—

^a Total parameters across all three base models plus Meta-MLP (4095 parameters).

^b Total sequential training time (sum of three base models plus Meta-MLP training of 1.2 min). With parallel base model training, the effective wall-clock time reduces to approximately 8.5 min.

monitoring systems such as the Yangtze River Hydrological Network). This inference speed is well within the latency requirements of real-time flood warning systems, which typically require forecasts to be generated within minutes of data availability.

Hardware requirements: The total parameter count of RGL-Stacking across all components is 318,850 (Table 8), including 4095 parameters for the Meta-MLP. This is substantially smaller than Transformer-based approaches (524,289 parameters) and can be deployed on standard CPU-based server infrastructure without requiring GPU acceleration during inference. The base models and Meta-MLP can be serialized (e.g., using PyTorch's TorchScript or ONNX format) and loaded into memory once, enabling sub-second predictions for any number of incoming data points. This makes RGL-Stacking deployable in existing hydrological information centers with commodity hardware, without specialized deep learning infrastructure.

Comparison with the Transformer baseline: It is noteworthy that the Transformer model, despite its higher computational cost (28.4 min training, 0.08 s inference, 524,289 parameters), achieves lower NSE than RGL-Stacking at all four stations (Table 5). This confirms that RGL-Stacking offers a superior accuracy-to-computation trade-off for the Yangtze River water level prediction task.

5.5. Practical implications for real-time river monitoring and flood management

The RGL-Stacking model has direct and immediate applicability to two critical operational domains: (1) the Yangtze River Navigation Channel management system for daily navigation safety, and (2) real-time flood early warning systems for the middle and lower Yangtze River basin.

Operational deployment pipeline for navigation channel management:

The Yangtze River Navigation Channel management system requires high-precision water level predictions, particularly during the dry season (December–March) when vessel draft limitations impose strict minimum water depth requirements. A practical deployment pipeline would operate as follows.

- Data ingestion:** Real-time daily water level measurements collected from gauging station telemetry systems (interfacing with the Yangtze River Hydrological Network at <http://www.cjh.com.cn>) are automatically preprocessed—quality control, gap filling, and normalization—using parameters established during training. This step requires no manual intervention under normal conditions.
- Forecast generation:** The preprocessed 16-day water level sequence is fed into the pre-trained, serialized RGL-Stacking model (three base models + Meta-MLP) to generate the next-day water level prediction. Inference requires approximately 0.05 s per station, enabling forecasts for all four stations to be generated within 0.2 s of data availability—well within the requirements of systems operating on daily update cycles.
- Decision support for navigation:** The predicted water levels are used to inform navigation scheduling, particularly regarding: (i) vessel

draft limitation advisories during low-water periods; (ii) dynamic adjustment of shipping lane depth profiles; and (iii) early notification to shipping operators when predicted levels approach minimum navigable depth thresholds. The MAE of 0.0595–0.0866 m achieved by RGL-Stacking is well within the precision requirements of navigation management systems (typically ± 0.1 – 0.2 m).

Deployment for flood early warning:

For flood management applications, the model's demonstrated superiority during the flood season (days 170–220, mid-June to early August) is particularly relevant. A flood early warning deployment would integrate RGL-Stacking as follows.

- a) Flood threshold monitoring: Predicted water levels are continuously compared against predefined warning thresholds (e.g., the Yangtze River's standard flood warning levels of 40.3 m at Hankou and 14.5 m at Nanjing). When predicted levels approach or exceed these thresholds, automated alerts are triggered for relevant emergency management authorities.
- b) Cascade warning for multi-station networks: Given the approximately 1200 km spatial extent of the four monitored stations (from Yichang downstream of the Three Gorges Dam to Nanjing in the tidal reach), predictions at upstream stations (Yichang, Shashi) can be used to issue advance warnings for downstream stations (Hankou, Nanjing), leveraging the known flood wave travel times of approximately 1–3 days between stations.
- c) Model updating for non-stationary conditions: The Three Gorges Dam's operational strategies have evolved since its completion in 2003, and climate change is expected to alter long-term hydrological patterns. To maintain forecast accuracy under these evolving conditions, the model can be periodically retrained (recommended quarterly) with newly accumulated data. The low training cost (approximately 8.5 min with parallel training) makes this operationally feasible without disrupting ongoing monitoring services.
- d) Integration with existing hydrological information systems: The RGL-Stacking model's lightweight inference footprint (0.05 s/station, deployable on CPU) enables seamless integration into existing hydrological information center infrastructure, such as the National Flood Control and Drought Relief Headquarters' real-time monitoring platform, without requiring hardware upgrades.

Limitations for real-time flood management: It should be noted that the current model provides only one-day-ahead point predictions without uncertainty quantification. For operational flood risk management, probabilistic forecasts with prediction intervals are more appropriate (addressed in Section 5.6f). Additionally, the model's performance during extreme flood events beyond the historical training range (2003–2010) remains untested, and operational deployment should include fallback procedures for anomalous conditions.

5.6. Limitations and future work

Despite the promising results, several limitations should be acknowledged.

- a) Univariate input: The current model uses only historical water level data as input. Incorporating exogenous variables such as precipitation, upstream reservoir discharge, and temperature could improve prediction accuracy, particularly during extreme events driven by meteorological forcing. Even simple derived features such as the day-to-day water level change ($\Delta x_t = x_t - x_{t-1}$) or calendar features (day-of-year encoded as sine/cosine) could provide additional predictive information without requiring external data sources.
- b) Single prediction horizon: The current model focuses on one-day-ahead prediction. Extension to multi-day-ahead forecasting (e.g., 7, 14, 30 days) would enhance its operational utility for medium-term

planning but would require careful consideration of error accumulation (for recursive strategies) or training separate models for each horizon (for direct strategies).

- c) Geographic scope: The model is validated only on four stations in the middle and lower Yangtze River. Transferability to other river basins with different hydrological regimes (e.g., arid-region rivers, snowmelt-dominated rivers) remains to be tested.
- d) Stationarity assumption: The model assumes that statistical patterns learned from the 2003–2010 training period remain representative of the test period. Long-term non-stationarity driven by climate change, evolving Three Gorges Dam operation policies, or upstream land use changes could degrade performance over time, necessitating periodic retraining.
- e) Limited data volume: The relatively short study period (2610 days) constrains the amount of training data. Future studies with longer records (e.g., 15–20 years) could enable more robust training and the exploration of longer input windows.
- f) Uncertainty quantification: The current model provides only point predictions. For risk-based flood management, probabilistic forecasts with prediction intervals would be more useful. Monte Carlo Dropout [41] or ensemble spread from multiple random seeds could be explored.
- g) Meta-learner simplicity: The Meta-MLP receives only three scalar predictions as input. Using the base models' hidden state vectors as meta-features could substantially increase the information available to the meta-learner, potentially yielding further improvements.

Future research will focus on: (i) incorporating exogenous variables (precipitation, reservoir discharge) as additional inputs; (ii) extending to multi-day-ahead prediction using direct multi-step strategies; (iii) testing on additional river basins to assess geographic generalizability; (iv) exploring attention-based meta-learners; (v) developing uncertainty quantification capabilities; and (vi) comparing with more recent state-of-the-art methods including PatchTST [42] and TimesNet [43].

6. Conclusions

This study proposed the RGL-Stacking model, a heterogeneous stacking ensemble deep learning model that combines RNN, GRU, and LSTM base learners with an MLP meta-learner for water level prediction in the Yangtze River. The key findings are summarized as follows.

- a) RGL-Stacking achieves consistently superior performance across all four stations. The model attains MAE values below 0.1 m (0.0866, 0.0683, 0.0595, and 0.0563 m at Yichang, Shashi, Hankou, and Nanjing, respectively) and NSE values of 0.998, 0.999, 0.999, and 0.998, outperforming individual RNN, GRU, LSTM, SVM, and Transformer models.
- b) The ensemble successfully addresses the station-dependent performance variability of individual models. While each single model shows varying accuracy across stations (RNN best at Hankou, GRU best at Shashi, LSTM best at Yichang and Nanjing), RGL-Stacking maintains consistently high accuracy without requiring station-specific model selection.
- c) Separate evaluations during dry and flood seasons confirm the model's robustness. RGL-Stacking consistently provides the best prediction accuracy during both low-water periods (critical for navigation safety) and high-water periods (critical for flood warning), demonstrating its adaptability to diverse hydrological conditions.
- d) Ablation experiments and residual correlation analysis verify the complementarity of base learners. The moderate inter-model residual correlations (mean 0.53–0.71) confirm that the three architecturally distinct base models make different types of errors, providing the diversity necessary for effective ensemble integration.

- e) The model achieves these improvements with manageable computational overhead (approximately $2.1 \times$ training time relative to a single LSTM), making it practical for operational deployment in the Yangtze River navigation channel management system.

The RGL-Stacking model offers a valuable methodological contribution for water level prediction in complex river systems. Its ability to integrate the complementary strengths of diverse recurrent architectures through nonlinear meta-learning provides a principled and effective approach for improving prediction robustness, offering practical support for water resource management, navigation safety, and flood control in the Yangtze River basin.

CRedit authorship contribution statement

Youwei Li: Writing – original draft, Funding acquisition, Formal analysis. **Xueting Lei:** Visualization, Software. **Bing Wang:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Kun Tan:** Methodology, Data curation. **Zhaobing Jiang:** Writing – original draft, Methodology. **Zhiyuan Ren:** Writing – review & editing, Visualization.

Disclosures

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of competing interest

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Data availability

Data underlying the results presented in this paper have been deposited in Zenodo (accession number: 10.5281/zenodo.10434269).

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